

Roadmap to Enhance the Census Tool for Grant Support:

A Research and Strategy Guide for Implementation

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1. Introduction

The Census Bureau initiative to support novice data users applying for federal grants through Grants.gov has established a tool that curates the most relevant statistics for grant applications. As the federal government allocates billions annually through more than 900 grant programs, state and local government agencies, non-profits, and tribal organizations (SLaNT users) often grant writers struggle with navigating complex data requirements. Although the tool has already shown its value, our group has done further research to assess gaps in the tool and how the tool could better help certain personas: such as Miguel (community leader), Divya (state partner), Marcus (city manager), and Sarah (NGO grant writer). This research paper presents a detailed roadmap for enhancing the Census tool in three key areas: grant assistance, functionality, and data visualization. Each enhancement is mapped against user needs, implementation strategies, timelines, and potential risks.

2. Enhancement Area I: Grant Assistance - Predicting Best Metrics for Grants

2.1 Summary of Enhancement The tool will feature an intelligent recommendation system that suggests the most relevant metrics based on the content of a specific grant opportunity. This feature will utilize natural language processing (NLP) to parse grant descriptions and match them to relevant data domains and metrics.

2.2 Benefits

1. Accelerates the grant writing process by reducing the time users spend searching for and validating relevant data. This process is particularly beneficial for novice users with limited data experience or under tight submission deadlines.
2. Enhances the quality and relevance of applications by ensuring that selected metrics are directly aligned with the language, objectives, and evaluation criteria of specific grant opportunities.
3. Reduces user burden in identifying which statistics to include. By selecting a curated list of metrics that align with the grant's themes, the tool minimizes confusion, eliminates guesswork, and ensures the even users with limited data experience can incorporate impactful, grant-appropriate statistics into their proposals.

2.3 Persona-Based Impact

- **Miguel** (community leader) will no longer need to search manually for relevant statistics, saving significant time.
- **Divya** (state partner) can ensure her applications are aligned with diverse, cross-sectoral needs.
- **Marcus** (city manager) will improve application precision without needing extensive data literacy.
- **Sarah** (NGO grant writer) will write more compelling data-backed narratives with less stress.

2.4 Implementation Strategy

- Build a keyword-to-tag mapping between common grant terms and the top metrics. This foundational step involves creating a structured mapping system that links frequently used phrases or themes from grant applications. The mapping should rely on a curated glossary based on prior grant language analysis, domain-specific jargon, and expert tagging. Each keyword will be assigned tags representing relevant data domains (e.g., Health, Housing, Transportation), which then point to applicable Census metrics. This step creates the rule-based backbone for metric recommendation and must be maintained regularly to stay current with evolving grant terminology.
- Use NLP to extract key themes from grant descriptions. Once the keyword-metric relationships are established, natural language processing (NLP) techniques will be used to analyze the language within a grant's description. This includes parsing the title, purpose, objectives, and eligibility criteria. NLP will then identify dominant themes through a mix of term frequency analysis, phrase clustering, and semantic interpretation.
- Present a ranked list of recommended metrics with exportable justifications. Based on the extracted themes, the tool will display a prioritized list of metrics most relevant to the grant opportunity. Each recommendation will include a brief justification explaining why it was selected, helping users understand its relevance

2.5 Timeline

- **4-6 weeks** for MVP using **keyword mapping** – this is the process of linking specific words or phrases (keywords) commonly found in grant descriptions (like “housing,” “mental health,” or “infrastructure”) to correspond to statistical metrics or tags from

the dataset. This helps the system automatically identify and recommend the most relevant data points for a given grant application.

- **8-10 weeks** if implementing a full **machine learning model** with training data. This approach would go beyond keyword mapping by using training examples of past grant descriptions and their matched metrics to predict the most relevant statistics.

2.6 Risks

- Inaccurate recommendations due to vague or misleading grant descriptions – when grant descriptions use ambiguous or overly general language, the system may struggle to extract clear keywords or themes. For example, a grant that simply references "community support" without specifying the focus area (e.g., housing, mental health, youth development) may result in irrelevant or too broad metric suggestions. This can lead to applicants selecting less impactful data or failing to demonstrate the true alignment between their community's needs and the grant's intent. Ensuring users have the option to refine the context or select priority themes could help mitigate this risk.
- Overreliance on automated suggestions by users – if users begin to treat the tool's recommended metrics as definitive without applying context or domain judgment, they may submit applications that lack depth, or alignment with their priorities.
- Resource needs for maintaining up-to-date keyword-stat mappings – as the language used in grant solicitations evolves and new funding priorities emerge, the keywords used to match grants with relevant statistics may also shift. Maintaining an accurate and effective keyword-stat mapping system will require regular reviews, updates to the keyword database

3. Enhancement Area II: Visualization - Drillable Mapping Feature

3.1 Summary of Enhancement A multi-level geographic visualization system will be developed, enabling users to drill down from state to county to city levels. This will offer a more nuanced understanding of spatial data relevant to grant proposals.

3.2 Benefits

1. Facilitates localized data storytelling by allowing users to visualize and communicate insights that are specific to the geographic scale most relevant to their communities. For instance, applicants can highlight disparities in economic indicators between neighboring cities or illustrate the impact of a program at the neighborhood level. This kind of storytelling adds emotional and contextual depth to proposals

2. Enables users to demonstrate regional disparities and target impact areas by visually highlighting where certain challenges or strengths are more concentrated across different geographic levels. For example, a user could show how unemployment or housing instability is significantly higher in one county compared to neighboring ones, providing compelling justification for funding.
3. Enhances credibility and specificity of grant narratives. Rather than relying on generalized or national-level statistics, applicants can use drill-down mapping to identify highly localized metrics, such as poverty rates or infrastructure conditions within specific city blocks or counties. This level of granularity lends credibility to the application by demonstrating a deep understanding of the local context.

3.3 Persona-Based Impact

- **Divya** (state partner) can isolate county or city-specific trends within her statewide programs.
- **Marcus** (city manager) can evaluate neighborhood-level infrastructure or economic data.
- **Miguel** (community leader) **and Sarah** (NGO grant writer) can showcase how specific communities fare against broader geographies.

3.4 Implementation Strategy

- **Integrate Mapbox or ESRI for advanced map rendering to support high-quality, interactive geographic visualizations.** Mapbox offers flexible, web-friendly mapping APIs and libraries that allow for smooth drill-down experiences and dynamic rendering across geographic layers. ESRI provides powerful GIS capabilities and advanced analytics tools that can support complex spatial queries, custom layers, and integration. These platforms enable responsive map performance, visually rich user interfaces, and compatibility with mobile and desktop devices. Selecting between them will depend on licensing availability, user interface design needs, and the team's familiarity with the respective software.
- **Utilize Census Bureau's geographic API for dynamic loading of shapefiles,** allowing the tool to fetch and display geographic boundary data such as census tracts, counties, and cities in real time. This approach ensures the visualizations are always current and aligned with official Census-defined boundaries. It also enhances flexibility by enabling the system to load different geographies on demand based on user interaction, rather than hardcoding static shapefiles into the platform. Additionally, leveraging the API supports sustainable development

practices by reducing storage requirements and simplifying updates when geographic definitions change.

- Provide toggles for users to change levels and export custom maps.

3.5 Timeline

- 6-8 weeks for a basic drill-down prototype – This initial version would include interactive functionality allowing users to navigate from higher geographic levels (such as states) to lower levels (such as counties or cities), with a user interface designed to support intuitive zooming and filtering. It would also incorporate dynamic rendering of geographic shapes and data overlays for selected metrics, offering a foundational layer of geographic interactivity. The focus during this phase would be on achieving reliable performance, geographic accuracy, and a clean, accessible design before advancing to more complex features.
- 10-12 weeks for full implementation with all geographic levels and responsive design. This phase would build upon the prototype to include seamless transitions between geographic levels, comprehensive visualizations for all supported metrics, and an interface that is fully responsive across devices (desktop, tablet, mobile). The development would also account for accessibility features like keyboard navigation and screen reader compatibility, as well as optimized loading for complex map layers. This level of refinement ensures the map component is both technically robust and usable by the broad spectrum of users described by the personas' needs.

3.6 Risks

- Geographic data granularity may vary across domains. Some datasets provide detailed statistics at each census level, others may only be available at broader levels such as county, state, or national aggregates. For example, employment data might be available down to the zip code level, while healthcare access data may only be available at the county level. These inconsistencies in the data can affect the ability of users to generate precise visualizations. This potentially limits the usefulness of the tool in certain geographic areas.
- Performance issues with rendering high-resolution geographic layers. As geographic data becomes more granular, such as down to census tracts, blocks, or neighborhood segments, the file sizes and complexity of the visualizations increase substantially. To address this, future teams may need to employ strategies such as tile caching, progressive rendering, and limiting the number of visible layers at once, while balancing responsiveness.

4. Enhancement Area III: Functionality - Easy Comparison Between Two Areas

4.1 Summary of Enhancement This enhancement introduces a comparison mode that allows users to view side-by-side statistics and visualizations for two user-defined geographic areas. Users will be able to select any two geographic units—such as city vs. city, county vs. county, or city vs. state—and directly compare relevant Census metrics like population characteristics, poverty rates, education levels, and infrastructure indicators. The comparison will be displayed using synchronized charts, tables, and narrative descriptions to ensure interpretability. This feature aims to improve the ability to benchmark needs, identify gaps, and tailor funding justifications to clearly communicate disparities or similarities between areas. Additionally, users can export these comparisons for inclusion in grant documents.

4.2 Benefits

1. **Supports compelling comparisons in funding justification.** Grant reviewers often expect applicants to contextualize their community's needs relative to others. The ability to easily compare key metrics side-by-side provides a clear and data-driven foundation for articulating disparities or demonstrating progress. This helps applicants make a more persuasive case for why funding is needed in one area over another.
2. **Provides users with tools to benchmark and highlight disparities.** This feature helps users compare their community's statistics with those of similar or more developed areas. For example, someone applying for a housing grant could compare local housing conditions with a nearby city that has better housing outcomes. These comparisons make it easier to identify where help is needed most and give grant reviewers a clearer picture of why funding is important. It also helps communities set goals by showing what success could look like in similar places.
3. **Simplifies cross-jurisdictional application needs.** Many grant applications involve multi region partnerships. This comparison feature enables users to gather and present consistent, comparable statistics across different geographies. This streamlines data sourcing and ensures alignment in multi-jurisdictional proposals.

4.3 Persona-Based Impact

- **Sarah** (NGO grant writer) can highlight inequality between her agency's service area and wealthier regions.
- **Miguel** (community leader) can justify community needs by comparing his area to better-funded regions.

- **Divya** (state partner) can demonstrate regional gaps within her state.
- **Marcus** (city manager) can align his city's metrics with neighboring cities.

4.4 Implementation Strategy

- Introduce a dual-area selector tool in the UI. Users should be able to intuitively search and select two geographic units from a map or dropdown interface.
- Use synchronized charts and metric cards for comparison. Comparison results should include aligned visual metric card allowing for an intuitive side-by-side review.
- Integrate contextual summaries. Auto-generate brief narrative statements that explain key differences between areas, enhancing usability for lower-literacy users or those crafting narratives quickly.

4.5 Timeline

- 3-5 weeks for MVP assuming existing metrics and visual layers. This version would enable users to select two areas and display a small subset of commonly requested metrics (e.g., income, population, education).
- 6-8 weeks for more advanced visual comparison and UI testing. This would include a more robust interface, full metric selection capabilities, export functions, and feedback from user testing. Refinements would also ensure accessibility and consistency in comparative displays.

4.6 Risks

- Confusion when comparing different geography types (e.g., city vs. county). Users may draw misleading conclusions if the geographic scales are too different. The interface should include clear labels, tooltips, and warnings to guide valid comparisons.
- Data inconsistencies between selected regions. Not all datasets are available at every geographic level or for all areas. The system must gracefully handle missing data, alerting users and offering nearest alternatives.
- UI overload if too many comparison options are shown. If users are presented with too many metrics at once, it may overwhelm or confuse them. To prevent this, we would recommend just allowing two comparisons.

5. Broader Considerations and Recommendations

5.1 Technical Enablers

- Secure licenses for ESRI/Mapbox. These technologies are essential for rendering dynamic maps and geographic visualizations. ESRI offers sophisticated GIS capabilities, while Mapbox provides flexible, customizable maps suited for web platforms.
- Incorporate user-centered design principles through iterative feedback cycles. Engage a diverse group of users—reflecting the range of personas—to test features regularly, identify usability issues, and provide feedback. Iterative improvements should focus on reducing cognitive load and supporting intuitive navigation.
- Maintain real-time API integration to ensure tool sustainability. By continuously syncing with the Census API, the tool can reflect updated data as it becomes available without requiring manual refreshes. This ensures data accuracy and long-term relevance.

5.2 Potential Risks Across the Project

- Overcomplication for low-tech users. The personas indicate limited data experience, so interfaces must be designed with simplicity, guidance, and fail-safe defaults. Avoid jargon and ensure that first-time users can accomplish basic tasks.
- Mismatched expectations if users assume tool will cover non-Census grants or private funders. Since the tool is based exclusively on Census data and focused on federal funding, the UI and help content must reinforce this boundary to avoid confusion or disappointment.

6. Conclusion

Enhancing the Census grant tool with predictive metric recommendations, drillable maps, and comparison features addresses major barriers for novice users such as Miguel, Divya, Marcus, and Sarah. These features collectively empower users to access, interpret, and apply authoritative statistics effectively in their grant applications. The roadmap presented in this paper outlines clear, actionable strategies to implement each feature while balancing usability, accessibility, and sustainability. With thoughtful design and stakeholder engagement, these enhancements will significantly expand the impact and value of the Census tool in the federal grants ecosystem.

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